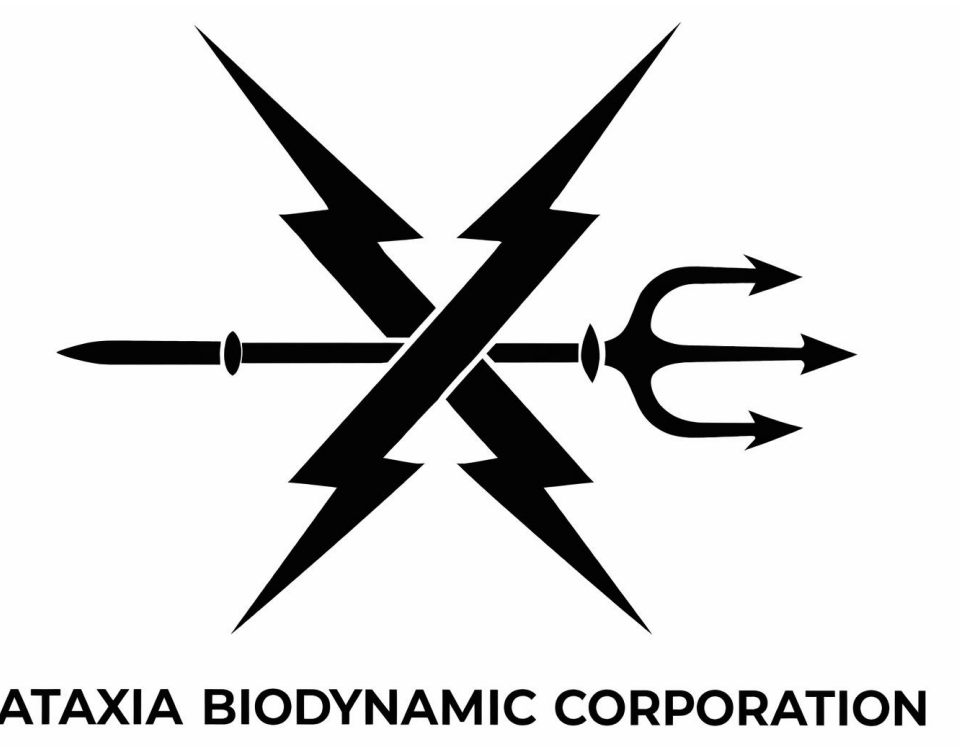


# Self-Defense Device (Zeus)

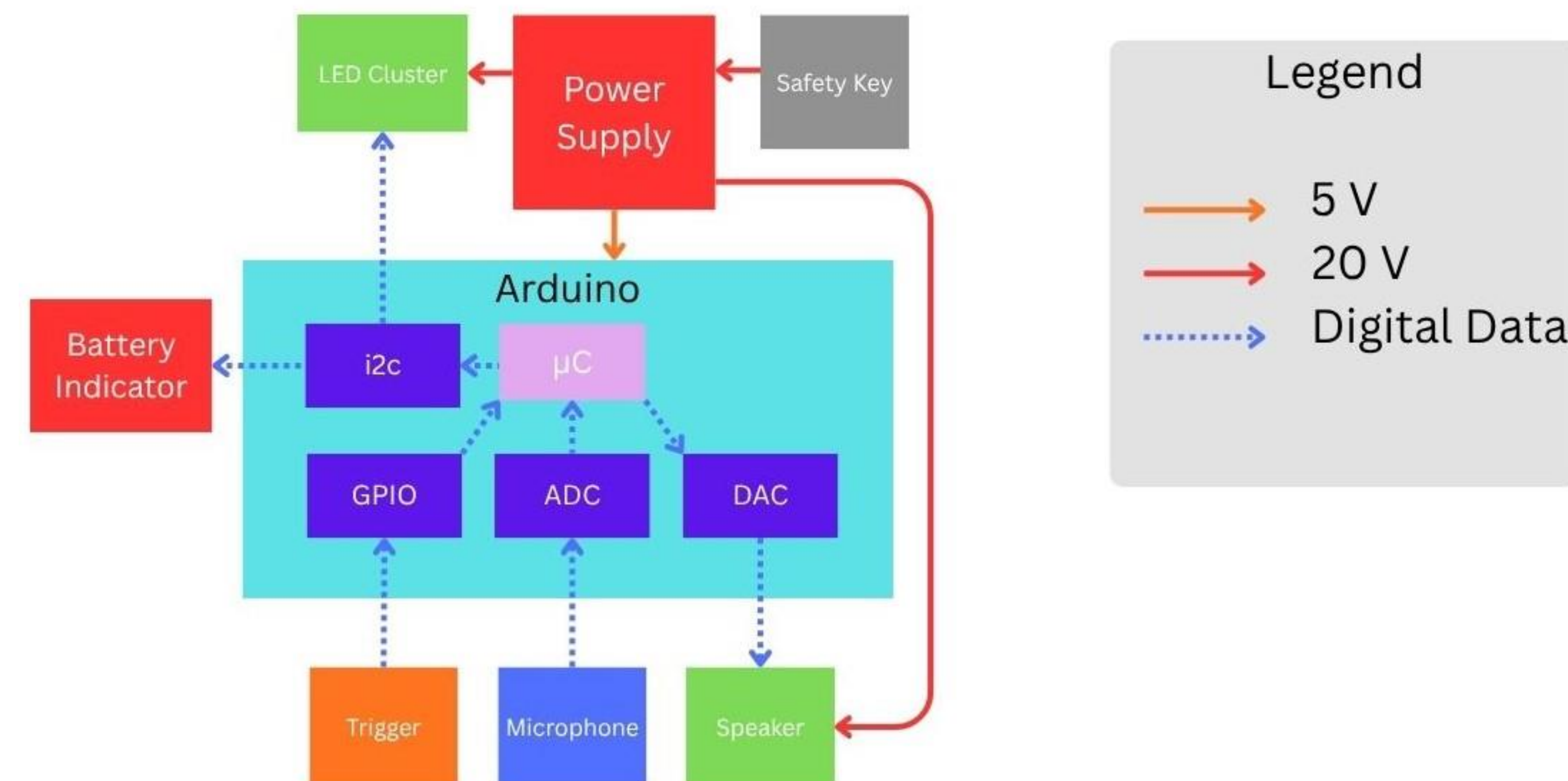
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## Abstract

Zeus is a handheld self-defense device designed to protect school staff and civilians during active threat situations. It uses high-intensity sound and strobe lighting to temporarily disorient an attacker from a safe distance, buying critical time until law enforcement arrives. This device addresses the lack of effective, less-lethal safety tools. It focuses on a solution that is both easily accessible and compliant with regulations. Zeus is designed to be easy to use in high-stress situations, combining ergonomic controls, built-in safety features, and regulated output to ensure effectiveness without causing permanent harm to the aggressor or bystanders

## Hardware Diagram



## Competing Products

|                                 | Area Denial                                                                                   | Ease of Use                                                                                         | Range                                                                     | Effectiveness                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Zeus Device (\$1k)              | <b>Great</b> , the sound can be projected across a large space with little need for accuracy. | <b>Great</b> , the device has a single trigger and is identifiable as a gun, making it easy to use. | <b>Decent</b> , however, the sound is limited to 30 ft.                   | <b>Great</b> , the device affects the target with LEDs and high dB, disrupting senses.               |
| BeamQ Laser Dazzler (\$270)     | <b>Poor</b> , the dazzler requires pin point accuracy.                                        | <b>Decent</b> , the dazzler has a single button and is easy to use, but requires great accuracy.    | <b>Great</b> 1k-3k meters                                                 | <b>Decent</b> , this device is great if accurate, but is limited in sensory scope.                   |
| Axon-Taser Pulse 2 (\$399)      | <b>Poor</b> , the taser effects only one target at a time.                                    | <b>Great</b> , the taser is identifiable as a gun and is easy to operate.                           | <b>Poor</b> , 15 feet; close to midrange                                  | <b>Decent</b> , probe must make direct contact with skin so clothing can protect the target.         |
| Mace Pepper Gel (\$15.99)       | <b>Decent</b> , it can slow or fully stop a target before they enter an area.                 | <b>Great</b> , the canister has a flip top with a button to fire the gel.                           | <b>Poor</b> , 12-15 ft for small canister or 25-30 ft for large canister. | <b>Great</b> , it's difficult to defend against as it affects sight and causes a body reaction.      |
| Fenix Pro Flashlight (\$119.95) | <b>Poor</b> , does not provide true area denial.                                              | <b>Decent</b> , simple to use with a switch for 6 modes and a button to turn on and off.            | <b>Great</b> , 380 meters                                                 | <b>Poor</b> , the simplicity and wide focal length makes this device produce limited disorientation. |

## Need for Product

School staff currently lack effective and practical tools to respond during active threat situations. Existing self-defense options introduce safety risks, require close proximity, or are not permitted in school environments. This leaves a critical gap in protection while law enforcement arrives. These few minutes pose the highest risk to the safety of staff and students. A compliant, less-lethal, and user-friendly solution is necessary to enhance safety and response capability

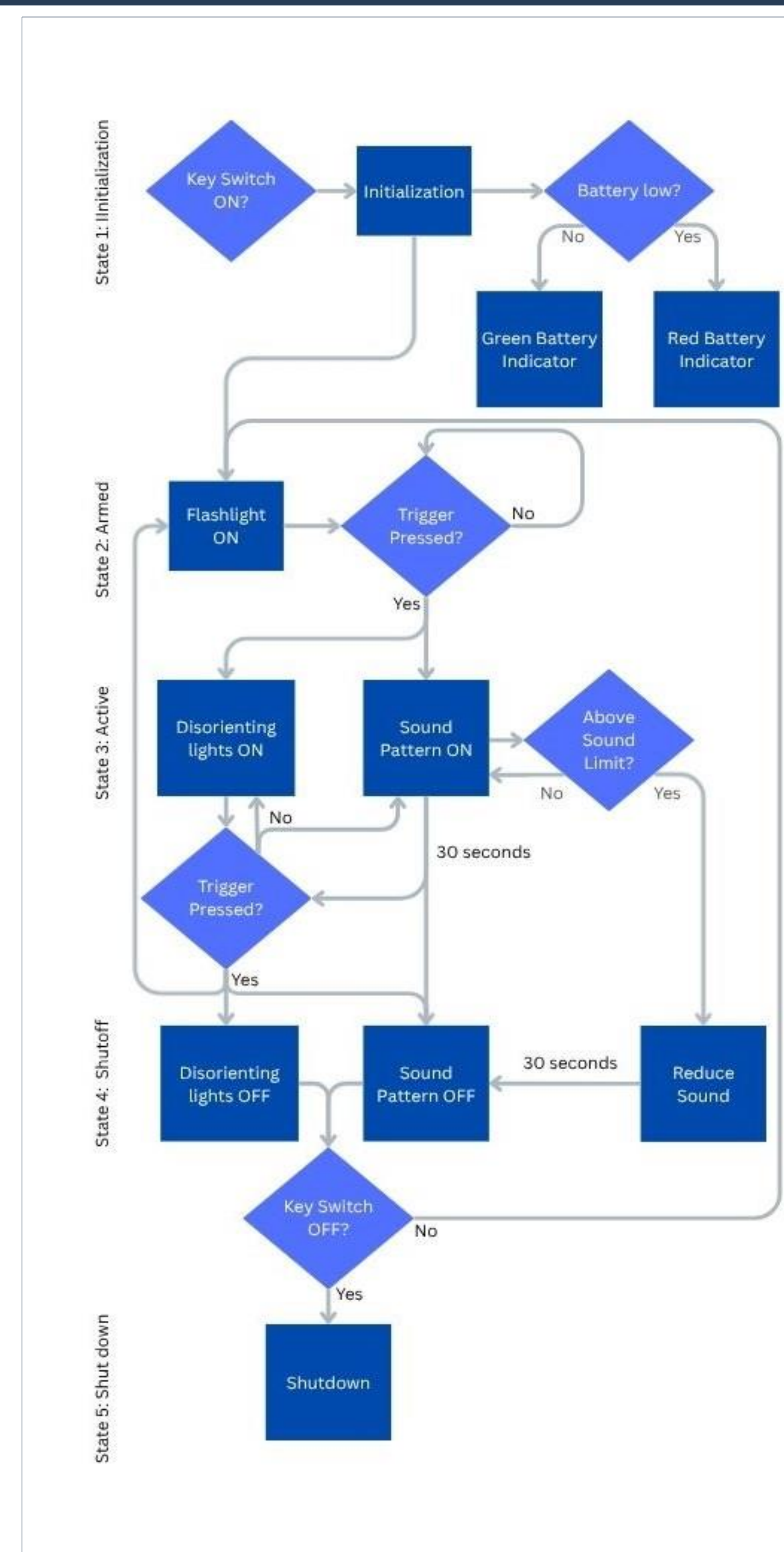


## Main Components

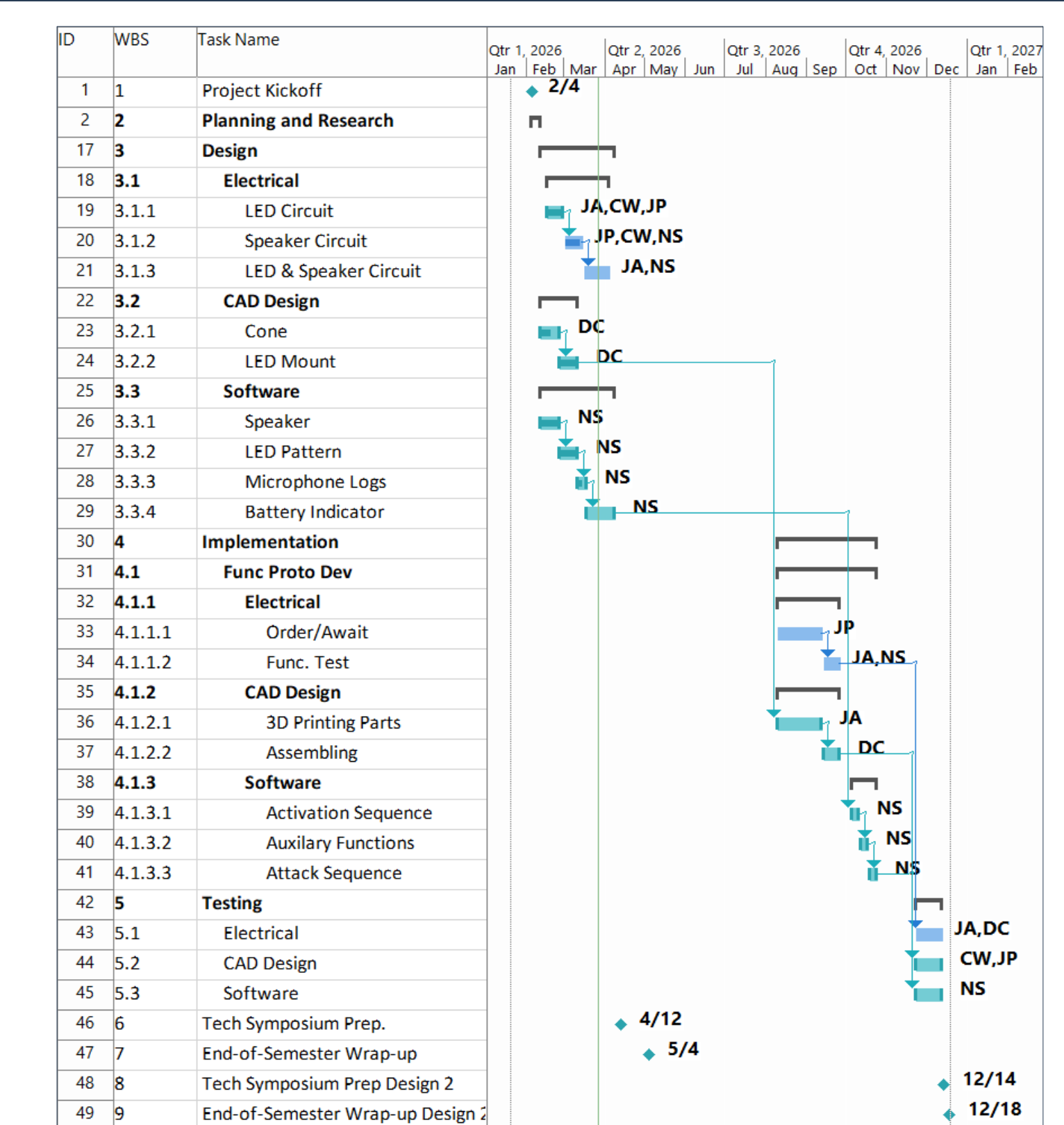
- Trigger (1\$)
- Microcontroller (15\$)
- Speaker Unit (100\$)
- White/Green LED Clusters (300\$)
- Battery Pack (150\$)
- Microphone (10\$)
- Battery Indicator (5\$)



## Software Flowchart



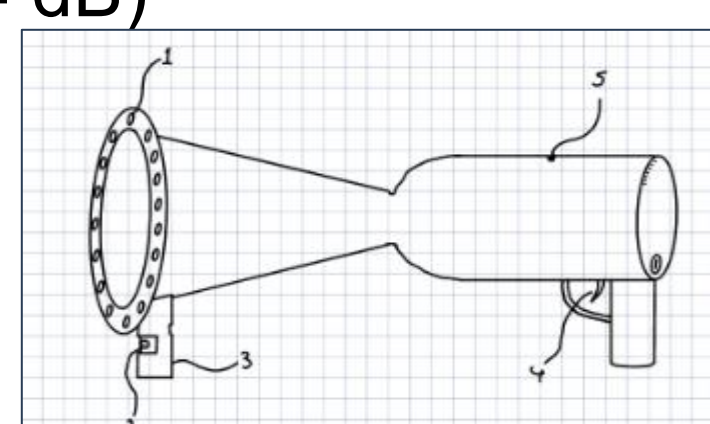
## Work Breakdown Structure



## Product Requirements

The Zeus system is designed to provide a less-lethal method of defense during active threat situations using directional high-intensity sound and strobe lighting to disorient an aggressor from a safe distance. It must operate reliably in high-stress environments, remain easy to use, comply with institutional and legal regulations, and ensure output levels are controlled to prevent permanent harm to the user and bystanders.

- **Directional** high-decibel sound output (100–134 dB)
- **High-intensity** strobe lighting for disruption
- **Simple trigger-based** activation
- **Portable** battery-powered design
- **Built-in safety monitoring**



## Acknowledgements

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